

12 (a) (i)	time for number of atoms/nuclei or activity to be reduced to one half	M1	
	reference to (number of...) original nuclide/single isotope		
	or		
	reference to half of original value/initial activity	A1	[2]
(ii)	$A = A_0 \exp(-\lambda t)$ <u>and</u> either $t = t_{1/2}$, $A = \frac{1}{2}A_0$ or $\frac{1}{2}A_0 = A_0 \exp(-\lambda t_{1/2})$	M1	
	so $\ln 2 = \lambda t_{1/2}$ (and $\ln 2 = 0.693$), hence $0.693 = \lambda t_{1/2}$	A1	[2]
(b)	$A = \lambda N$		
	$N = 200 / (2.1 \times 10^{-6})$	C1	
	$= 9.52 \times 10^7$	C1	
	mass = $(9.52 \times 10^7 \times 222 \times 10^{-3}) / (6.02 \times 10^{23})$		
	or		
	mass = $9.52 \times 10^7 \times 222 \times 1.66 \times 10^{-27}$	C1	
	$= 3.5 \times 10^{-17} \text{ kg}$	A1	[4]